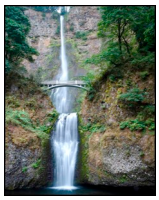
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A natural meditation setting improves compliance with mindfulness training

Freddie Lymeus[a,*], Per Lindberg[a], Terry Hartig[b]

[a] Department of Psychology, Uppsala University, Box 1225, SE-75142, Uppsala, Sweden
[b] Institute for Housing and Urban Research, Uppsala University, Box 514, SE-75120, Uppsala, Sweden

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ABSTRACT

The setting matters in meditation, but most research has neglected it. Many mindfulness-based health interventions emphasize effortful attention training exercises in sparsely furnished indoor settings. However, many beginners with attention regulation problems struggle with the exercises and drop out. In contrast, restoration skills training (ReST) – a five-week course set in a garden environment – builds on mindfulness practices adapted to draw on restorative processes stimulated effortlessly in nature contacts. Expecting that the ReST approach will facilitate the introduction to mindfulness, we compared drop-out and homework completion records from four rounds of ReST vs. conventional mindfulness training ($N = 139$). Randomly assigned ReST participants had lower drop-out and more sustained homework completion over the course weeks. Supporting the theoretical assumptions, higher restorative environmental qualities and state mindfulness mediated the compliance differences. The improved acceptability with ReST means that more people can enjoy the long-term benefits of establishing a meditation practice.

1. Introduction

Restoration skills training (ReST) is a five-week meditation-based course set in a natural environment (Lymeus, Lindberg, & Hartig, 2018). ReST was developed with two overarching aims. First, in contrast to conventional mindfulness training (CMT), the ReST practices should build a meditative state effortlessly on restorative experiences supported by the environment. Second, ReST should over time confer benefits at least comparable to those of CMT.

It follows from the first aim that the ReST approach should be more acceptable than CMT, and therefore help more people to establish a meditation practice. This article addresses that assumption by analyzing compliance records from a study comparing successive iterations of ReST with CMT. If compliance is higher with ReST, then more people could enjoy the eventual benefits mentioned in the second aim, which we address elsewhere (Lymeus et al., 2018; Lymeus et al., in preparation). In the following, we summarize the reasoning behind ReST, overview previous findings, and specify our hypotheses.

1.1. About restoration skills training

The ReST approach to meditation builds on exploration of sensory impressions and related experiences in a pleasantly interesting setting characterized by vegetation, water, and other natural features and processes (Lymeus et al., 2018). For a conception of mindfulness, the work with ReST has drawn mainly on the operational definition proposed by Bishop et al. (2004), describing mindfulness as attentiveness to present experience with qualities of curiosity and detachment (or decentering, in their terms), and on theory that defines different phases and states in mindfulness practice (e.g., Davidson & Kaszniak, 2015; Lutz, Jha, Dunne, & Saron, 2015). For the practice approach, the work took a starting point in the well-established Mindfulness-Based Stress Reduction (MBSR; Kabat-Zinn, 1990). ReST integrates those mindfulness concepts and practices with knowledge about restorative environments (Lymeus et al., 2018).

Reviews have shown that common mindfulness courses that involve class meetings and regular meditation over several weeks can produce long-term improvements in adaptive capabilities, reflected in enhanced performance on tests of selective and executive attention, reduced psychophysiological stress, increased positive affect, and more (e.g., Eberth & Sedlmeier, 2012; Tang, Holzel, & Posner, 2015). Repeated activation of neural attention networks in the meditation exercises presumably drives cognitive improvements in the early stages of training (e.g., Malinowski, 2013; Tang et al., 2015). The early stages mainly involve so-called focused-attention exercises, in which participants recruit cognitive control mechanisms to try to attend to a given target stimulus (e.g. sensations emerging with the breath), suppress mind-wandering, and patiently redirect attention when they become

* Corresponding author.
E-mail addresses: Freddie.Lymeus@psyk.uu.se (F. Lymeus), Per.Lindberg@psyk.uu.se (P. Lindberg), Terry.Hartig@ibf.uu.se (T. Hartig).

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distracted (Fox et al., 2016; Lutz et al., 2015). In contrast to such effortful exercises, ReST uses open-monitoring exercises in which participants use minimal effort to connect with external and internal aspects of experience (Lutz et al., 2015). Focused-attention and open-monitoring practice are both attended by relaxed alertness as reflected in increased alpha-band activity (Lomas, Ivtzan, & Fu, 2015); however, open-monitoring is associated with enhanced processing of sensory inputs and attentional coordination rather than control (Fox et al., 2016). Open-monitoring is commonly employed with more experienced meditators because beginners typically lack in ability to balance effortlessness with attentional stability (Lutz et al., 2015; Malinowski, 2013; Tang et al., 2015). The ReST approach works to overcome that inability not by effortful attention network training over several weeks but by engaging participants with pleasantly interesting natural stimuli that help regulate attention bottom-up and restore overworked attention regulation capabilities during meditation sessions (Lymeus et al., 2018; cf. Kaplan, 2001).

Restoration is the replenishment of an adaptive resource that has become depleted in efforts to meet demands, so that the associated functional capabilities are reinstated (Hartig, 2017). For example, a person working on an attention-demanding task in a busy open-plan office might initially perform well but over time come to experience fatigue and concentration problems (e.g., Jahncke, Hygge, Halin, Green, & Dimberg, 2011). If that person visits a natural setting, the experience of fatigue can more readily be relieved and attentional focus regained. Reviews have shown that nature contacts, more than comparable activities in indoor or urban public settings, can produce improvements in adaptive capabilities, reflected in enhanced performance on tests of selective and executive attention, reduced psychophysiological stress, increased positive affect, and more (e.g., Hartig, Mitchell, De Vries, & Frumkin, 2014; Stevenson, Schilhab, & Bentsen, 2018). In the terms of Kaplan and Kaplan's (1989; Kaplan, 1995) attention restoration theory, this is because natural settings support effortless bottom-up attentiveness to pleasantly interesting stimuli in the environment, known as fascination; and psychological distance from demanding tasks and mental routines, known as being away. Recent studies have developed our understanding of the involved mechanisms, considering how natural scenes require little cognitive effort to process (Grassini et al., 2019), reduce maladaptive rumination (Bratman, Hamilton, Hahn, Daily, & Gross, 2015), and provide gentle exogenous support for the attentional orienting network, thus facilitating switching between mind-wandering and external stimuli (Williams et al., 2018). As with mindfulness meditation, contacts with natural settings are also attended by increased alpha-band activity (Aspinall, Mavros, Coyne, & Roe, 2013; Grassini et al., 2019).

In developing ReST, we wanted to stimulate synergy and capitalize on complementarities between restorative nature experience and mindfulness meditation (Kaplan, 2001; Lymeus et al., 2018). For synergy, ReST draws on restorative processes sustained by the environment to support beginning meditators in effortless attentiveness to present experience, with fascinating natural features and processes stimulating curiosity and the psychological distance afforded by the environment easing detachment. Reciprocally, ReST uses open-monitoring practices to promote engagement with environmental experiences and minimize the disruptive influence of thoughts and other intrusions, thus bolstering and enhancing restorative processes. For complementarity, ReST combines the accessible short-term benefits of restorative nature experience with the longer-term benefits of regular training in skills that over time can enhance adaptation in many situations. Next, we overview previous findings from our studies comparing ReST with CMT.

1.2. ReST compared to CMT

Over four data collection rounds, we compared ReST, given in a botanic garden, with CMT which emphasized focused-attention practice in a sparsely furnished indoor setting. The participants were students with stress and/or concentration problems.

Lymeus et al. (2018) report outcomes on tests of selective and executive attention. On course weeks 1, 3, and 5, the participants completed tests before and after 90-min meditation classes (Study 1) or 20-min meditation sessions (Study 2). The ReST participants showed signs of restoration (i.e. performance improvement) with the meditation. Those restorative benefits increased over the course weeks, affirming that ReST both allows restorative processes and trains restoration skills over time. CMT participants initially showed some signs of restoration but then increasingly over the course weeks showed signs of effort (i.e. performance decrement) with the meditation. CMT apparently incurred increasing effort as participants gained skills in focused-attention practice.

Lymeus et al. (2018) also compared longer-term attention outcomes by looking at the trends over the course weeks in the assessments obtained before the meditation classes and sessions. The more effortless practice approach of ReST was attended by longer-term performance benefits comparable to those seen with CMT.

Lymeus et al. (in preparation) report outcomes on ratings of dispositional mindfulness, cognitive functioning, and chronic stress, obtained before and after the course and six months later. The ReST and CMT participants showed similar improvements which, for dispositional mindfulness and cognitive functioning, were sustained six months after both courses. These results affirm that ReST works as a mindfulness training course without relying on effortful meditation practices, producing long-term benefits in the studied psychological health domains comparable to those realized with a conventional course.

1.3. Aims and hypotheses

Health interventions are only effective if people in the target population will accept them. Acceptability can be measured as compliance, seen for example in how many participants complete the protocol and follow the recommendations. According to Sekhon, Cartwright, and Francis (2017), acceptability depends on the balance between perceptions of the effort and other costs of participation, one's capacity to meet those demands, and the anticipated or experienced benefits (as well as other aspects of less concern here).

We reasoned that while many beginning meditators may experience an initial honeymoon phase in which the practice is perceived as relatively easy and immediately beneficial, many will soon come to struggle with conventional mindfulness exercises. As the balance between the perceived burden and benefit shifts, compliance may dwindle. In fact, drop-out rates from MBSR and Mindfulness-Based Cognitive Therapy (MBCT; Segal, Williams, & Teasdale, 2002) frequently exceed 25% (cf. Abbott et al., 2014; Lamothe, Rondeau, Malboeuf-Hurtubise, Duval, & Sultan, 2016; Nam & Toneatto, 2016).

People with attention problems may be particularly unlikely to comply with conventional mindfulness courses. Studying formerly depressed people participating in MBCT, Crane and Williams (2010) found that drop-outs initially had more problems with intrusive negative thoughts, and greater decline in problem solving performance with a negative mood induction, than course completers with comparable emotional symptoms. Banerjee, Cavanagh, and Strauss (2018) showed that participants with more initial problems with intrusive negative thoughts had poorer compliance even if they formally completed an online mindfulness course. Lymeus, Lundgren, and Hartig (2017) reported that students with poorer initial attention test performance attended fewer MBSR classes and completed fewer homework exercises than participants with already stronger attention, independent of emotional symptoms.

The attentional demands incurred with CMT may therefore be too high for many of the individuals who have the most to gain from establishing a regular practice. ReST could be a more acceptable

introduction to mindfulness training. Building on these ideas, our aim was to compare compliance records from our research contrasting ReST against CMT. We also aimed to determine how theoretically relevant constructs from the mindfulness and restorative environments literatures could explain any compliance differences. We test these hypotheses:

1: Compared with CMT, ReST will be attended by higher compliance in the form of course completion (H1a) and homework completion (H1b).
2: Initial attention problems will have less influence with ReST than with CMT on course completion (H2a) and homework completion (H2b). To complement this interaction test, we also explore effects on compliance of initial dispositional mindfulness, cognitive functioning, and chronic stress.
3: Higher perceived restorative quality of the class setting and higher state mindfulness in the classes will mediate the differences between ReST and CMT in course completion (H3a) and homework completion (H3b).

2. Methods

2.1. Participants and procedures

2.1.1. Enrollment

We posted flyers on our university campus asking for volunteers for a study about mindfulness training. Participants had to study ≥75% of full time, experience stress or concentration problems, and have little or no meditation experience. A screening interview excluded those with relevant disorders (Dobkin, Irving, & Amar, 2012) or ongoing treatments. Participation was free. Participants could drop out at any time without facing further requests or consequences. They were, however, promised three cinema tickets if they completed the course and all measurements. The research was approved by the regional ethical review board (registration number: 2013/033) and adhered to the Declaration of Helsinki.

We enrolled different participants in four separate data collection rounds, planning to eventually pool data from different rounds as appropriate to test different hypotheses. For each round, however, we conducted separate power analyses with a particular view to the tests of attention administered before and after the classes or practice sessions (as subsequently reported in Lymeus et al., 2018). These analyses anticipated 25% drop-out, fairly high correlation among measurements ($r \approx .70$), and small to medium-sized Course type x Time (2 × 2) interactions. We concluded that 40 initial participants in a round should suffice for medium effects ($f \approx .25$) while 60 initial participants would allow detection of a more satisfactory range of effects ($f \approx .15$–$.20$). However, we also had practical constraints. In Round 1, we could only accommodate 20 course participants; in Round 2, we aimed for 40 but only recruited 31 within the allocated time-window; in Round 3, we aimed for at least 40 and recruited 49; in Round 4, we aimed for 60 and recruited 59. To show how well the sample sufficed for tests of the compliance outcomes of concern here, we present sensitivity analyses in the Results section (3) for the two central tests for Course type differences in compliance (H1a, H1b).

2.1.2. Assignment

Participants were stratified by gender and randomly assigned to ReST or CMT within each data collection round. Altogether, 159 participants were assigned. Seventeen withdrew before the first class. Three at some point revoked consent to handle their data. The available data therefore come from 139 participants: 68 who started ReST (44 women; median age = 24) and 71 who started CMT (49 women; median age = 23) (Fig. S1 shows details of participant flow). In Rounds 2–4, the participants were accommodated in multiple course groups of up to 12 participants. These met on different weekdays. Participants self-selected a group and could not switch during the course.

2.1.3. Course format

The 5-week courses had one 90-min class each week. They used brief exercise durations (15–30 min in-class, 15–20 min homework). The homework assignments involved a formal and an informal exercise each week, and instructions to practice with both on most days between classes. An experienced clinical psychologist and mindfulness instructor led the work and collaborated with different psychologists-in-training in providing the courses.

The classes included meditation as well as theoretical discussion and homework consultation. Each class introduced new concepts and exercises. In the homework consultations, the instructors gave a rationale and advice for establishing a regular practice habit. All course contents were based on a preliminary script to ensure their quality and consistency: contact the first author to discuss possibilities for using these scripts in replications.

2.1.4. ReST

The ReST classes were held in a botanic garden, mostly in a tropical greenhouse which in pilot studies had shown higher perceived restorative quality compared to different indoor campus settings (Kihlberg, 2012; Nordfeldt, 2012). The in-class and homework exercise instructions reflected the ReST approach; however, the homework instructions did not require meditation in natural settings because this could have introduced undue constraints. The rationale for regular practice emphasized the benefits of connecting with experiences in daily life.

2.1.5. CMT

The CMT classes were held in a university campus building, in small classrooms which, building on pilot studies (Kihlberg, 2012; Nordfeldt, 2012), were expected to be low in both restorative qualities and distractions. The in-class and homework exercise instructions were adapted from MBSR exercises. They emphasized focused-attention practice and contact with internal experiences. The rationale for regular practice emphasized the benefits of training the mind in being present.

2.2. Measures

2.2.1. Compliance records

We operationalized drop-out as having come to at least one class and then either giving notice of discontinuing or missing at least two sequential classes and not returning for subsequent classes. All participants were coded as either completer (1) or drop-out (0).

We obtained homework records for each course week. The participants registered completed homework for each day between the classes and handed in the registrations at the following class meeting. We summed the registrations for each of the four weeks (no new homework was assigned in the last class).

2.2.2. Predictors of compliance

We obtained measures of potential predictors of compliance before the courses started.

We measured attention performance with the Letter–Digit Substitution Test (LDST; van der Elst, van Boxtel, van Breukelen, & Jolles, 2006). LDST has previously shown predictive power regarding compliance with mindfulness training (Lymeus et al., 2017). The participants were instructed to quickly and accurately fill in digits in blank spaces corresponding to randomly assigned letters, guided by a code-key. Following a practice run, the score was the number of correct responses completed in 60 s ($M = 37.97$, $SD = 5.98$).

We measured dispositional mindfulness with the Five Facet Mindfulness Questionnaire (FFMQ; Baer, Smith, Hopkins, Krietemeyer, & Toney, 2006; Lilja et al., 2011). It has 29 statements about how often in the last month a person had different mindfulness experiences

(1 = never, 5 = always). Internal consistency (Cronbach's α) was .86. The score was the mean of all item responses (M = 3.03, SD = 0.43). Higher scores indicate higher dispositional mindfulness.

We measured general cognitive functioning with the Cognitive Failures Questionnaire (CFQ; Broadbent, Cooper, FitzGerald, & Parkes, 1982). It has 25 questions about how often in the last month a person made mistakes reflecting cognitive failures (0 = never, 4 = very often). Internal consistency was .87. The score was the mean of all item responses (M = 1.83, SD = 0.51). Higher scores indicate poorer cognitive functioning.

We measured chronic stress with the Perceived Stress Scale (PSS; Cohen, Kamarck, & Mermelstein, 1983). It has 14 questions about how often in the last month a person felt unable to handle the demands in their life (0 = never, 4 = very often). Internal consistency was .83. The score was the mean of all item responses (M = 1.95, SD = 0.49). Higher scores indicate higher chronic stress.

2.2.3. Mediators of compliance

We obtained measures of potential mediators of compliance at the end of the classes on course weeks 1, 3, and 5.

We measured perceptions of the restorative quality of the class setting with 10 items from the Perceived Restorativeness Scale (PRS; Hartig, Kaiser, & Bowler, 1997; Hartig, Korpela, Evans, & Gärling, 1997). The items are statements about the degree to which a person perceives the environment as supporting fascination and being away, respectively (0 = not at all, 10 = completely). For the three measurement occasions, internal consistency was .92, .93, and .95. The score was the mean of all item responses (week 1 M = 6.31, SD = 1.94; week 3 M = 6.55, SD = 1.99; week 5 M = 6.76, SD = 2.17). Higher scores indicate higher restorative quality.

We measured state mindfulness in the classes with the Toronto Mindfulness Scale (TMS; Lau et al., 2006). It has 13 statements about the degree to which a person experienced curiosity and decentering in meditation (0 = not at all, 4 = very much). For the three measurement occasions, internal consistency was .83, .87, and .89. The score was the mean of all item responses (week 1 M = 2.30, SD = 0.57; week 3 M = 2.41, SD = 0.60; week 5 M = 2.53, SD = 0.58). Higher scores indicate higher state mindfulness.

2.3. Design and statistical analyses

The participants were randomly assigned to ReST or CMT in four data collection rounds. The study thus had two between-subjects factors: Course type (ReST, CMT) and Round (1–4). The rounds had comparable design and measurements; however, randomization over the rounds was not possible.

Homework, PRS, and TMS data were obtained on multiple occasions. For missing scores, we carried forward the last value, avoiding replacement where ≥2 sequential scores were missing. We chose this simple imputation method to avert complications in predicting values for non-randomly missing data, given that the attenuation of the variability and trends should be minor when only single missing values in the series were replaced. In the moderation and mediation models, we used the individual average score (PRS_{MEAN}, TMS_{MEAN}) to represent general levels. We used individual ordinary least squares regression coefficients to represent the development over the course weeks (per Lorch & Myers, 1990; cf. Twisk, 2004), producing variables representing the intercepts: $Homework_{ij}$ (M = 7.68, SD = 3.29), PRS_{ij} (M = 6.42, SD = 1.87), TMS_{ij} (M = 2.35, SD = 0.57), and the slopes: $Homework_{bj}$ (M = −0.36, SD = 0.97), PRS_{bj} (M = 0.17, SD = 0.57), TMS_{bj} (M = 0.08, SD = 0.32).

To control for effects of Round, we included Course type, Round, and Course type x Round in analyses where appropriate, treating Round as multicategorical using indicator coding. In the mediation models, we allowed covariates: demographics (gender [1 = female, 0 = male], age [years]) and initial assessments (LDST, FFMQ, CFQ, PSS; entered one by one to limit parameter estimates/N).

For statistical inferences, we used a double-tailed $p < .05$ criterion and 95% confidence intervals (CI). We used the SPSS general linear model for RM-ANOVA and the SPSS PROCESS macro (Hayes, 2017) for moderation (model 1) and mediation analyses (model 6). PROCESS produced bias-corrected CI's and heteroscedasticity-consistent (HC3) standard errors based on 10 000 bootstrap samples.

2.3.1. Hypothesis 1

For a Course type difference in course completion (H1a), we used Chi-square analysis. We used PROCESS to probe moderation by Round.

For a Course type difference in homework completion (H1b), we used RM-ANOVA (Course type x Week) to contrast the average and trends over time. We used PROCESS with $Homework_{bj}$ to probe moderation by Round.

2.3.2. Hypothesis 2

For prediction by initial attention problems of course completion (H2a) and $Homework_{bj}$ (H2b), we used PROCESS with LDST as predictor and Course type as moderator. We also explored prediction by FFMQ, CFQ, and PSS moderated by Course type.

2.3.3. Hypothesis 3

For mediation of course completion by experiential qualities in the classes (H3a), we used PROCESS with Course type predicting the outcome through a serial path including PRS_{MEAN} and TMS_{MEAN}. The order reflected reasoning that restorative qualities of the setting would support participants in meditation, facilitating state mindfulness and course completion. We checked the final model with the first PRS and TMS scores obtained from each participant to ensure that any difference between course completers and drop-outs was not confounded by data missingness in the later weeks among the drop-outs, and with the mediators entered in reverse order to rule out alternative explanations.

For mediation of $Homework_{bj}$ by experiential qualities in the classes (H3b), we used PROCESS with Course type predicting the outcome through a serial path first including the intercepts PRS_{ij}, TMS_{ij}, and $Homework_{ij}$. Their order reflected reasoning that restorative qualities of the setting would support the participants in meditation, facilitating state mindfulness and homework practice from the course start. The serial path then included the slopes TMS_{bj} and PRS_{bj}. Their order reflected reasoning that improvement in mindfulness skills over the course weeks would support the participants in engaging with restorative qualities of the setting during meditation. Increasing perceived restorativeness during meditation would, in turn, support maintenance of the practice habit. We checked the final model with the mediators entered in alternative orders to rule out alternative explanations.

3. Results

3.1. Compliance differences between the course types (H1)

Before the course started, nine participants withdrew from ReST and eight from CMT: withdrawal was independent of Course type ($\chi^2[1] = 0.10$, $p = .754$, $\varphi = .025$). The remaining sample of 139 gave a test sensitivity of $\varphi \approx .23$ ($1 - \beta = .80$) for the 2 × 2 Chi-square analysis of course completion. Of the 68 participants who started ReST, seven (10%) dropped out. Of the 71 participants who started CMT, 19 (27%) dropped out: a significant difference ($\chi^2[1] = 6.19$, $p = .013$, $\varphi = .211$). In line with H1a, course completion was therefore higher with ReST. The Course type difference varied marginally over the rounds (likelihood ratio test for interaction: $\chi^2[3] = 6.71$, $p = .082$): we therefore checked for effects of Round also in the subsequent analyses of course completion.

Looking at the 122 participants who provided homework data for the first course week, we saw that ReST and CMT participants had similar initial homework completion ($F[1,120] = 1.89$, $p = .172$, $\eta_p^2 =$

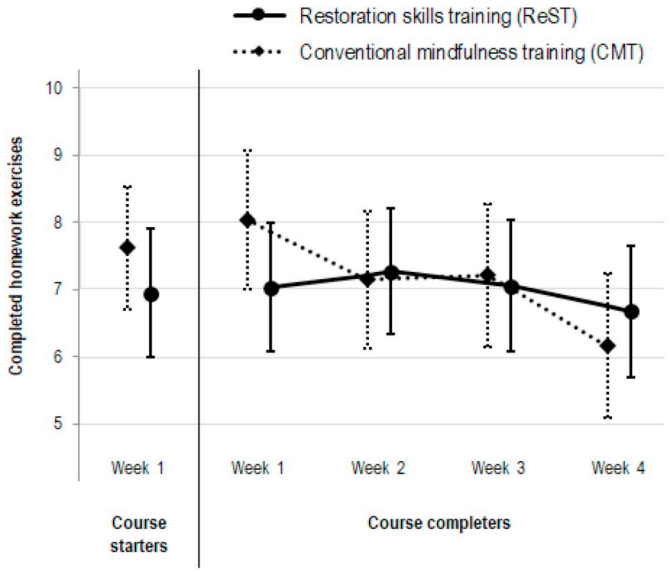


Fig. 1. Trends in the reported number of completed homework assignments across the course weeks for restoration skills training (ReST) and conventional mindfulness training (CMT) participants. Error bars show 95% confidence intervals.
Note. Some course starters and course completers did not provide sufficient homework data to be included in the analyses. Data were available for 122 course starters (ReST $n = 61$, CMT $n = 61$) and 102 course completers (ReST $n = 57$, CMT $n = 45$).

.015). Turning to the 113 course completers, 102 provided sufficient homework data to be included in the analyses of trends over the course weeks. This sample gave the RM-ANOVA a test sensitivity of $\eta^2 \approx .012$ ($1 - \beta = .80$) for the Course type (2) x Time (4) interaction. The analysis showed that on average, both ReST and CMT participants completed about seven homework exercises per course week (ReST $M = 7.03$ [CI: 6.22, 7.83], CMT $M = 7.29$ [CI: 6.38, 8.20]; Table S1 gives descriptive statistics per course week and round). However, a pronounced linear trend of decline over the course weeks differed between the ReST and CMT participants (Week linear trend: $F[1,100] = 16.30$, $p < .001$, $\eta_p^2 = .140$; Course type x Week linear trend: $F[1,100] = 4.79$, $p = .031$, $\eta_p^2 = .046$). Fig. 1 shows that the ReST completers – a larger proportion of the initial sample – sustained a steady homework practice from week to week (Homework$_{bj}$ $M = -0.13$; CI: -0.39, 0.13). With CMT, however, the more decimated sample who would eventually end up completing the course had somewhat higher initial homework completion than the ReST participants. They then dropped from week to week (Homework$_{bj}$ $M = -0.56$; CI: -0.84, −0.27) to a somewhat lower homework completion rate. In line with H1b, homework completion was therefore more sustained with ReST. However, the Course type difference varied over the rounds ($\Delta R^2 = .070$, $F[3,94] = 3.51$, $p = .018$): we therefore checked for effects of Round also in the subsequent analyses of homework.

3.2. Initial assessments as predictors of compliance, with moderation by course type (H2)

The initial LDST score did not interact with Course type to predict either course completion (interaction term coefficient = 0.011, $SE = 0.087$, $p = .895$) or Homework$_{bj}$ ($\Delta R^2 = .005$, $F[1,99] = 0.40$, interaction term coefficient = −0.022, $SE = 0.035$, $p = .530$): we cannot confirm H2a or H2b. In fact, neither LDST, FFMQ, CFQ, nor PSS scores predicted compliance reliably in either course type (see Table S2). Including Round as a covariate did not change the conclusions.

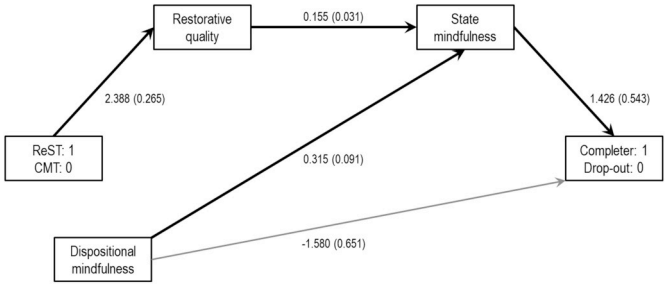


Fig. 2. Simplified model showing the difference in course completion between the restoration skills training (ReST) and conventional mindfulness training (CMT) courses mediated in serial through restorative quality (Perceived Restorativeness Scale) and state mindfulness (Toronto Mindfulness Scale) measured in connection with the meditation classes. Initial dispositional mindfulness (Five Facet Mindfulness Questionnaire), and demographics (gender, age; not shown) were included as covariates.
Note. Coefficients are unstandardized. Parentheses contain heteroscedasticity consistent (HC3) standard errors. Nonsignificant paths are omitted for clarity.

3.3. Mediation of compliance by experiential qualities in the classes (H3)

For course completion, the model including serial mediation through PRS$_{MEAN}$ and TMS$_{MEAN}$ fit the data best when we included demographics and initial FFMQ score as covariates (Final step Log Likelihood = 21.33, Nagelkerke pseudo-$R^2 = .238$, $p = .002$; see Fig. 2 and Table 1; Table S3 gives the statistics for tests without covariates). The serial path through PRS$_{MEAN}$ and TMS$_{MEAN}$ was significant (0.529, $SE = 0.294$; CI: 0.070, 1.178). The simple indirect paths were nonsignificant (through PRS$_{MEAN}$: −0.184, $SE = 0.520$; CI: −1.218, 0.845; through TMS$_{MEAN}$: −0.231, $SE = 0.186$; CI: −0.709, 0.009). The direct effect of Course type was also nonsignificant (1.251, $SE = 0.671$; CI: −0.064, 2.567). The model was not substantively changed by including Round, CFQ, or PSS as covariates or by using the first PRS and TMS ratings instead of the means. The serial path was not significant when we entered TMS$_{MEAN}$ before PRS$_{MEAN}$ (−0.023, $SE = 0.076$; CI: −0.228, 0.093): the effect of PRS$_{MEAN}$ on course completion was mediated through TMS$_{MEAN}$, but not vice versa. In line with H3a, the advantage for ReST over CMT in course completion could therefore be attributed in part to higher restorative quality in the setting facilitating the meditation.

For Homework$_{bj}$ the model included serial mediation through the intercepts PRS$_{ij}$, TMS$_{ij}$, and Homework$_{ij}$, followed by the slopes TMS$_{bj}$ and PRS$_{bj}$. Of the 102 participants who provided sufficient homework data, 99 also provided sufficient PRS and TMS data. The model held well on the whole (Final step $R^2 = .236$, MSE = 0.786, $F[6,92] = 5.30$, $p < .001$; see Fig. 3 and Table 2). Homework$_{ij}$ was independent of Course type, PRS$_{ij}$, and TMS$_{ij}$; however, we retained it because it contributed to prediction of TMS$_{bj}$, PRS$_{bj}$, and Homework$_{bj}$. The serial path through PRS$_{ij}$, TMS$_{ij}$, and PRS$_{bj}$ was significant (0.033, $SE = 0.029$; CI: 0.001, 0.111), as was the simple indirect path through PRS$_{bj}$ (0.152, $SE = 0.094$; CI: 0.009, 0.373). The intercepts expectedly also had negative relationships with their respective slopes, resulting in negative serial paths through PRS$_{ij}$ and PRS$_{bj}$ (−0.085, $SE = 0.057$; CI: −0.224, −0.003) and through PRS$_{ij}$, TMS$_{ij}$, TMS$_{bj}$, and PRS$_{bj}$ (−0.038, $SE = 0.027$; CI: −0.106, −0.003). Importantly, the relationship between TMS$_{bj}$ and PRS$_{bj}$ was positive. With the mediators, the direct effect of Course type was nonsignificant (0.084, $SE = 0.231$; CI: −0.374, 0.542). The model was not substantively changed by including Round, demographics, or initial assessments as covariates. The serial path through PRS$_{ij}$, TMS$_{ij}$, and PRS$_{bj}$ turned negative when we entered TMS$_{ij}$ before PRS$_{ij}$ (−0.012, $SE = 0.010$; CI: −0.039, 0.000), leaving only the serial path through TMS$_{ij}$ and PRS$_{bj}$ to convey some of the positive effect of ReST on homework completion (0.047, $SE = 0.038$; CI: 0.002, 0.144). None of the mediation paths, serial or simple, were

Table 1
Results from the successive steps in testing for the difference in course completion between the restoration skills training (ReST) and conventional mindfulness training (CMT) courses as mediated in serial through restorative quality (Perceived Restorativeness Scale; PRS) and state mindfulness (Toronto Mindfulness Scale; TMS) entered as the mean of scores obtained in connection with the meditation classes on course weeks 1, 3, and 5.

Step 1 DV = PRS_{MEAN}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
$R = .620$, $R^2 = .385$, MSE = 2.359, $F(4,132) = 20.57$, $p < .001$	Constant	5.396	1.025	3.369	7.423
	Course type	2.388	0.265	1.865	2.912
	Gender	0.161	0.263	−0.359	0.681
	Age	−0.018	0.022	−0.062	0.025
	FFMQ	0.069	0.295	−0.514	0.652
Step 2 DV = TMS_{MEAN}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
$R = .558$, $R^2 = .311$, MSE = .203, $F(5,131) = 8.01$, $p < .001$	Constant	0.227	0.371	−0.507	0.962
	Course type	−0.162	0.085	−0.331	0.007
	Gender	0.104	0.086	−0.065	0.274
	Age	0.009	0.005	−0.002	0.019
	FFMQ	0.315	0.091	0.136	0.494
	PRS_{MEAN}	0.155	0.031	0.094	0.216
Step 3 DV = Course completion	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
Nagelk. $R^2 = .238$, LL = 21.33, -2LL = 105.81, $p = .002$	Constant	2.823	2.451	−1.981	7.628
	Course type	1.251	0.671	−0.064	2.567
	Gender	−1.100	0.597	−2.269	0.069
	Age	0.044	0.057	−0.068	0.157
	FFMQ	−1.580	0.651	−2.857	−0.303
	PRS_{MEAN}	−0.077	0.171	−0.412	0.258
	TMS_{MEAN}	1.426	0.543	0.362	2.491

Note. Course type = ReST (1) vs. CMT (0), Gender = woman (1) vs. man (0), FFMQ = Five Facet Mindfulness Questionnaire (scale 1–5). PRS_{MEAN} and TMS_{MEAN} (scale 0–10 and 0–5, respectively) are the individual means of the ratings obtained in connection with the ReST and CMT classes on weeks 1, 3, and 5 of the course. Coefficients are unstandardized. Standard errors (*SE*) are heteroscedasticity consistent (HC3). LLCI and ULCI are the lower and upper limits of 95% confidence intervals.

significant when we instead entered PRS_{bj} before TMS_{bj} as the final steps in the model. In line with H3b, the more sustained homework completion rate with ReST could therefore be attributed in part to a more positive trend of development over the course weeks in perceived restorativeness during meditation. This trend was driven by high initial restorative quality supporting state mindfulness, high initial homework completion supporting a positive trend of development over the course weeks in state mindfulness, and a positive direct effect of Course type.

4. Discussion

ReST can be a more acceptable introduction to mindfulness training for people with stress or concentration problems than the conventional approach. ReST had higher course completion (H1a) and better establishment of a regular meditation practice (H1b). Our analyses also support the proposed mechanisms behind these differences: the perceived restorativeness of the setting supports beginning meditators in their practice, facilitating the meditation and helping them complete

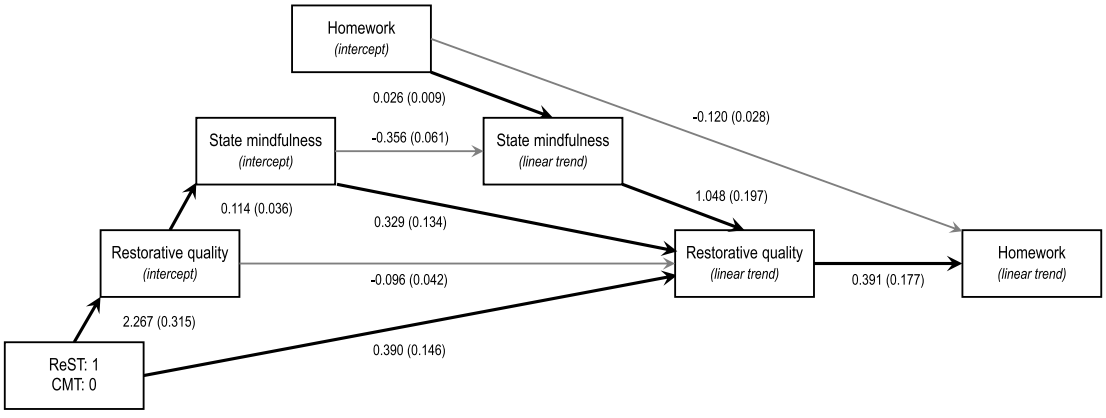


Fig. 3. Simplified model showing the difference in homework completion across the course weeks between the restoration skills training (ReST) and conventional mindfulness training (CMT) courses mediated in serial through restorative quality (Perceived Restorativeness Scale) and state mindfulness (Toronto Mindfulness Scale) measured in connection with the meditation classes. Restorative quality, state mindfulness and homework completion are represented by ordinary least squares regression intercepts and slopes calculated for each participant based on data obtained across the course weeks.
Note. Coefficients are unstandardized. Parentheses contain heteroscedasticity consistent (HC3) standard errors. Nonsignificant paths are omitted for clarity.

Table 2
Results for the successive steps in testing for the difference in homework completion across the course weeks ($Homework_{bj}$) between the restoration skills training (ReST) and conventional mindfulness training (CMT) participants as mediated in serial through regression intercepts ($_{ij}$) representing initial values and slopes ($_{bj}$) representing change across the course weeks in restorative quality (Perceived Restorativeness Scale; PRS) and state mindfulness (Toronto Mindfulness Scale; TMS).

Step 1 DV = PRS_{ij}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R = .598, R^2 = .358, MSE = 2.323, $F(1,97)$ = 51.81, p < .001	Constant	5.148	0.245	4.663	5.634
	Course type	2.267	0.315	1.642	2.892
Step 2 DV = TMS_{ij}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R = .439, R^2 = .193, MSE = 0.272, $F(2,96)$ = 9.04, p < .001	Constant	1.543	0.213	1.121	1.965
	Course type	0.109	0.123	−0.136	0.354
	PRS_{ij}	0.114	0.036	0.042	0.186
Step 3 DV = $Homework_{ij}$	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R = .188, R^2 = .035, MSE = 11.047, $F(3,95)$ = 1.07, p = .366	Constant	6.717	1.742	3.259	10.174
	Course type	−1.529	0.888	−3.293	0.234
	PRS_{ij}	0.307	0.254	−0.197	0.812
	TMS_{ij}	−0.073	0.605	−1.274	1.128
Step 4 DV = TMS_{bj}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R= .638, R^2 = .407, MSE = 0.069, $F(4,94)$ = 10.97, p < .001	Constant	0.551	0.135	0.283	0.819
	Course type	−0.043	0.072	−0.186	0.100
	PRS_{ij}	0.029	0.022	−0.016	0.074
	TMS_{ij}	−0.356	0.061	−0.476	−0.235
	$Homework_{ij}$	0.026	0.009	0.008	0.044
Step 5 DV = PRS_{bj}	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R = .547, R^2 = .299, MSE = 0.251, $F(5,93)$ = 8.68, p < .001	Constant	−0.355	0.325	−0.999	0.290
	Course type	0.390	0.146	0.100	0.679
	PRS_{ij}	−0.096	0.042	−0.179	−0.012
	TMS_{ij}	0.329	0.134	0.063	0.596
	$Homework_{ij}$	0.011	0.016	−0.022	0.043
	TMS_{bj}	1.048	0.197	0.657	1.439
Step 6 DV = $Homework_{bj}$	**Predictors**	**Coefficients**	**SE**	**LLCI**	**ULCI**
R = .486, R^2 = .236, MSE = 0.786, $F(6,92)$ = 5.30, p < .001	Constant	−0.331	0.485	−1.293	0.632
	Course type	0.084	0.231	−0.374	0.542
	PRS_{ij}	0.045	0.061	−0.075	0.166
	TMS_{ij}	0.206	0.212	−0.214	0.626
	$Homework_{ij}$	−0.120	0.028	−0.175	−0.064
	TMS_{bj}	−0.092	0.325	−0.738	0.554
	PRS_{bj}	0.391	0.177	0.038	0.743

Note. Course type = ReST (1) vs. CMT (0). The observed range for the continuous variables was: PRS_{ij}: 0.77, 9.85; for TMS_{ij}: 0.56, 4.01; $Homework_{ij}$: 0.00, 15.60; TMS_{bj}: −0.93, 1.20; PRS_{bj}: −1.00, 1.40; $Homework_{bj}$: −2.70, 2.30. Coefficients are unstandardized. Standard errors (*SE*) are heteroscedasticity consistent (HC3). LLCI and ULCI are the lower and upper limits of 95% confidence intervals.

the course (H3a). For those who completed the course, not only fixed aspects of the setting but also the gradual acquisition of skills in engaging with restorative qualities in the environment during the meditation helped them maintain a regular practice habit (H3b). These results fit with the findings reported in Lymeus et al. (2018) that meditation with the ReST approach can help restore attentional capabilities while CMT exercise depletes them. Furthermore, findings reported in Lymeus et al. (in preparation, 2018) show that the ease of ReST does not compromise the long-term benefits of beginning mindfulness training. With higher compliance, more people with stress and concentration problems can enjoy those benefits.

We could not confirm that initial attention problems predicted compliance less strongly with ReST than with CMT (H2a, H2b). Actually, none of the initial assessments reliably predicted course completion or change over the weeks in homework completion in either course type, possibly because the CMT in this research used a briefer course and exercise format than most common mindfulness courses (cf. Lymeus et al., 2017).

4.1. Strengths and limitations

Contrasting two formally comparable, feasible approaches to mindfulness training for participants who could have much to gain from the training gives pragmatic value to the conclusions. The compliance outcomes are based on behavioral records which show expected relationships with the self-reported mediators. The results are also largely coherent with our predictions and with results presented elsewhere. Beside these strengths, we see two limitations and one design feature that require consideration.

First, the data were obtained in four data collection rounds between which participants could not be randomized. We pooled the data, checked and controlled for effects of Round, and could isolate the effects of interest. However, uncontrolled factors associated with Round (e.g., seasonality, history, social processes) apparently contributed to extraneous variation in our data.

Second, our CMT course used exercises adapted from MBSR and the

drop-out rate was comparable to figures reported for MBSR and MBCT. Still, the conclusions cannot be directly generalized to these or other mindfulness courses that differ in format and contents.

Regarding our design, some might consider it a weakness that it conflates course type with setting instead of testing both ReST and CMT in both natural and indoor settings to factor out practice and setting effects. However, with conventional focused-attention practice targeting internal experiences, a rich and fascinating setting is probably more of a nuisance than a support for many beginning meditators. Similarly, the ReST practices that we built in and on a restorative natural setting would likely fail in attaining support and meaning in an environment selected to minimize external stimulation. We sought to compare ReST with a feasible conventional alternative, and saw no sufficient reason to include conditions that would be unlikely to benefit the participants. We thus argue that the design compares two bona fide, environmentally congruent approaches to meditation. Our mediation analyses indicate how restorative quality and state mindfulness worked together in producing the compliance benefits of ReST; however, the design precludes conclusions about the relative importance of the environment and other aspects of the practice approach.

4.2. Future directions

We have specified connections between restorative environments research and meditation science which provide grounds for further developments at the intersection of the fields. Meditation skills and practices matter for perceptions of the environment, facilitating connection with its restorative qualities. The setting also matters for meditation. We build on theory and experience in assuming that mindfulness practices must be harmonized with the processes supported by the environment to achieve the restorative effects seen with ReST. However, further research is needed to show how the different adaptations of the practice approach in ReST contribute to its outcomes, and whether some of these adaptations could also work in some other settings.

The effectiveness of mindfulness training for people with stress and concentration problems depends on how well they can accept and complete the training. Restorative environments theories can guide understanding of the involved resource dynamics (see von Lindern, Lymeus, & Hartig, 2017). In considering these dynamics, not only attention and other psychological resources are relevant, but also environmental and social resources such as the availability of suitable settings, uninterrupted time, support from relevant others, and the ability to carry the direct and indirect monetary costs (cf. Birtwell, Williams, van Marwijk, Armitage, & Sheffield, 2018; Carlson, 2018).

Once meditators progress beyond an introductory course, continued development could require some degree of effortful training. Starting with ReST may facilitate later effortful training through early improvements in restoration skills and available attention resources, allowing more people to continue along a path of meditation.

More experienced meditators can choose a setting and a practice that fit with each other as well as with their current skills, needs, and aims. ReST was developed in the garden setting for beginning meditators with stress and concentration problems, with the intention to meet their low initial skill-level, their needs to restore and build adaptive resources, and their aim to establish a meditation practice that could continue to benefit them over time. Other combinations of settings and practices may serve other groups and purposes. For guidance, researchers could look to the settings that meditators traditionally or spontaneously choose, and the specific stimuli that they use to frame and build their practice.

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Declarations of interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2019.05.008.

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